

HymeKo-Gömb — concentric cascade for signed-hypergraph learning

$$H_{\text{outer}}[v] = \text{concat}_m H_{\text{outer}}^{(m)}[v]$$

$$H_{\text{outer}}^{(m)}[v] = \text{mean}_{c:v \in c} \sum_i \text{proj}_{\sigma_i}(k_i^{(m)}) \cdot X[v_i]$$

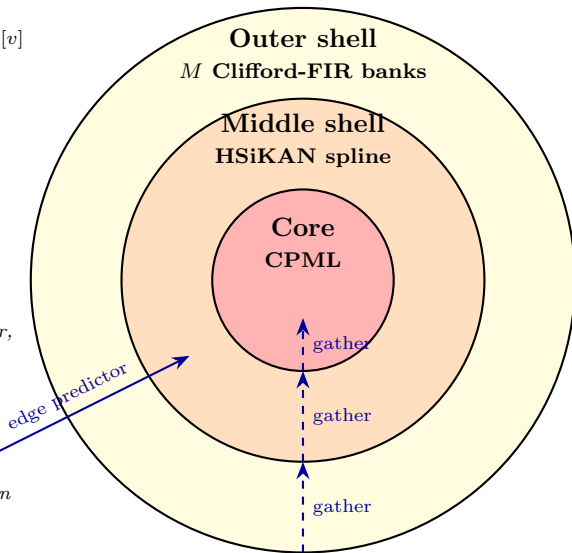
*M parallel FIR filters,
cheap and redundant*

$$H_{\text{middle}} = \text{HSiKAN}(H_{\text{outer}}, \mathcal{C}, \sigma)$$

*single Catmull-Rom layer,
adds nonlinearity*

$$H_{\text{core}} = \text{CPML}_{L=3}(H_{\text{middle}}, \dots)$$

tier-stratified compression



V1 (cortical periphery)
many oriented-edge
detectors in parallel

V4 (mid cortex)
curvature / shape
nonlinear transform

IT (deep cortex)
multi-scale identity
integration

Input: $(X, \mathcal{C}, \sigma)$